Is Your Cup Full? A Personal Invitation to Geofinitism

Why finity may be too much for those who have a cup full of piping hot infinity!

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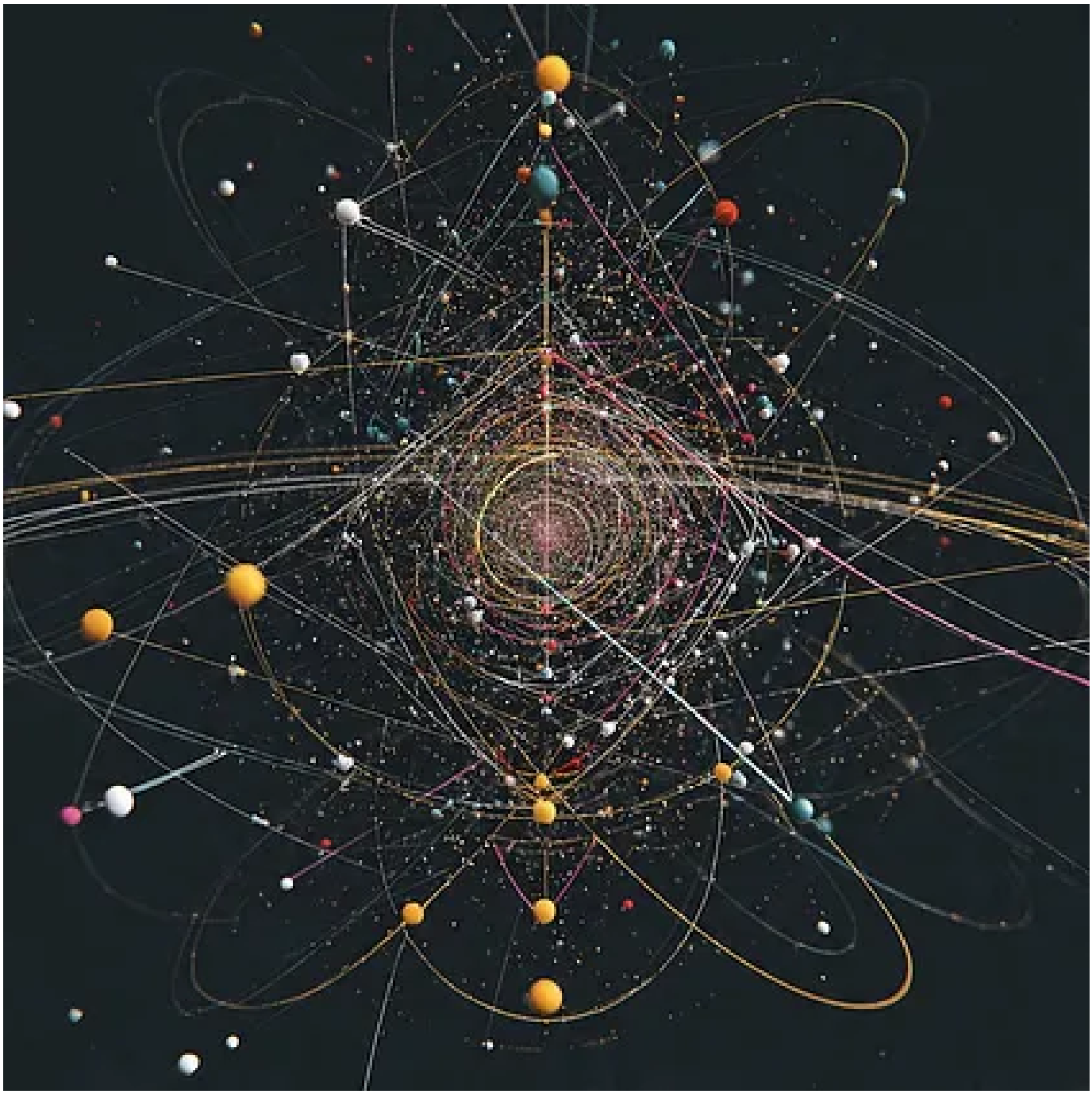
I've spent decades immersed in classical physics, mathematics, and engineering. My career was built in biomedical engineering: with a PhD, and 25 years of medical engineering research in the UK's health service, numbers, measurements, and falsifiable models were my daily bread and butter.



Then, one day, I decided to run an experiment on myself: what happens if I take finite axioms seriously, really seriously, and try to build physical models from them? I had been contemplating the ideas for decades, but always putting them to the side. Why? Because they fell outside of everything I could imagine. So I started scribbling, and playing thought games and one day I started writing equations. A few months later I had programs written and experimental results. Everytime I pushed the equations; they resolved:

> *Those experiments became Geofinitism.*

Changing how we see the world is hard. Asking others to consider a radically alternative viewpoint is harder still. It demands not just reading, but a willingness to suspend existing

beliefs, to compare, and sometimes to let go of existing ideas.

Most of the time, we get incremental change. New ideas layer gently atop the old. But what happens when the new idea is so fundamentally different that the mind rejects the very direction of the words from the outset?

That's the challenge I face when trying to share the Philosophy of Geofinitism. I'm not here to convert you. I'm not asking you to abandon your sense of meaning. I'm simply writing these words and wondering: do they connect with anyone else? Can the meaning I've found be seen, felt, or even better, be measured by another mind?

> *This is because in my view, the only world we can truly know is the one revealed through finite symbols, language, and measurement. And all measurements have uncertainty!*

Consider three simple questions:

- Are the symbols you're reading right now finite?
- Are you measuring them in some way?
- Do your measurements carry uncertainty?

Now consider the opposite position, the one baked into almost all analytical mathematics:

- These symbols have no finite form.
- You are not truly measuring them.
- They are perfect — Platonic, infinite, exact, idealization living in some unknowable unmeasurable space.

Most of us live comfortably in the second view, we don't even notice the viewpoint itself, as we don't question the finiteness of the symbols themselves. We work with them but imagine the symbols as tools. However, for me, they exist and they place a limit on how we can think and truly represent the world.

> *It seems that infinity is accepted as a foundation, a bedrock truth. Calculus, real numbers, continuum mechanics. These tools all rest upon the ideas that symbols themselves are excluded from the results they purport to represent.*

But what if we took the finite view seriously? What if we insisted that only finite, measurable things belong in our models of reality? What if any appeal to infinity is, by definition,

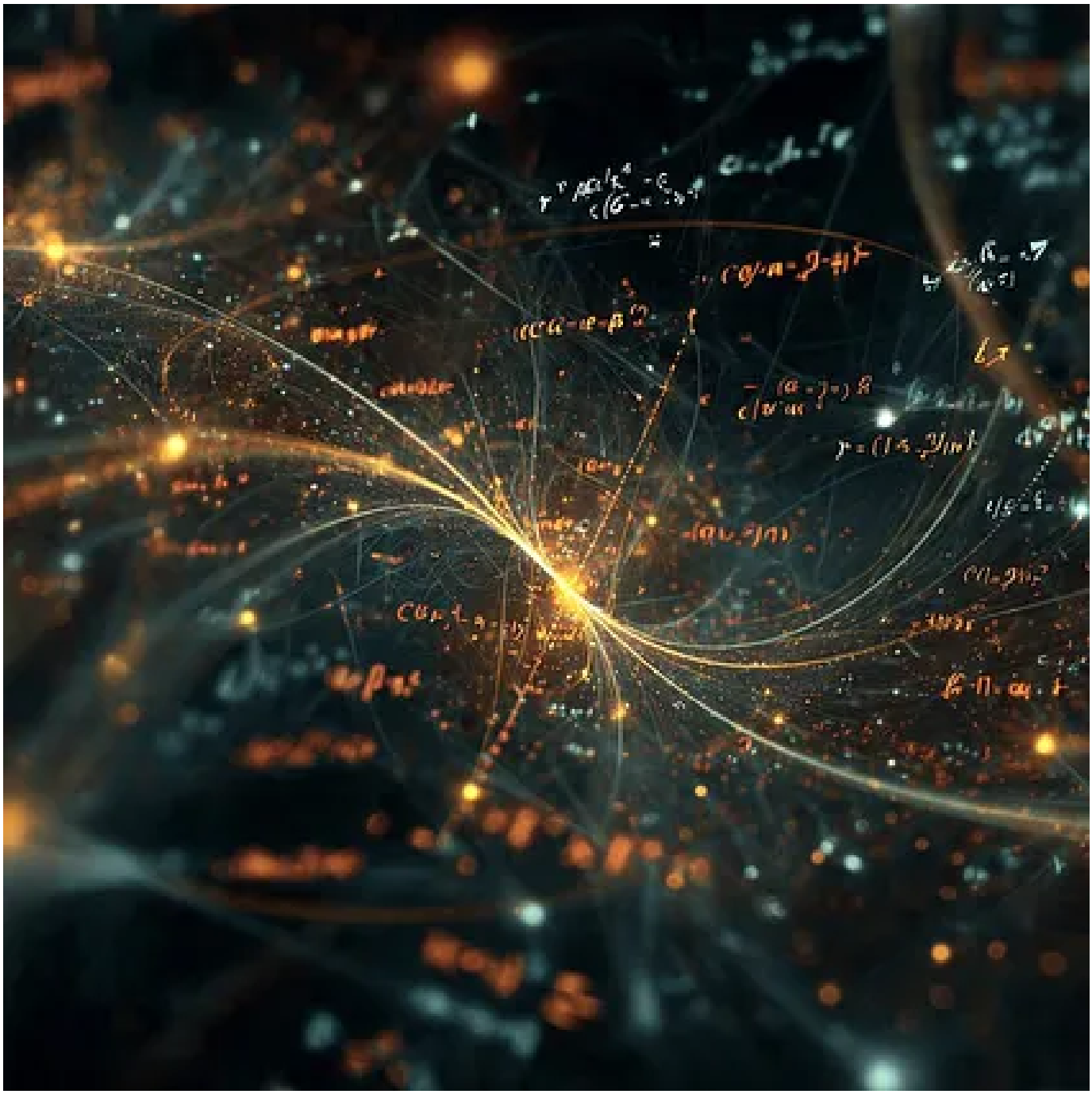
untestable ; and therefore unscientific?

This leap is huge and hard. Our attractors of language pull at us. Infinity is everywhere. What is beyond the edge? We can always add another one to one? Maybe, but every one is unique once it is instanced. And beyond the edge: that is just the space we have yet to measure. There will always be a space beyond; exactly because our measurements are finite. That infinity is just ‘beyond our measurements we have yet to make’.

But we want infinity to be a ‘thing’ to ‘exist’ to be more than a symbol, don’t we! I did, and do. But from a Geofinite perspective ‘infinity’ is just a symbol, a pointer to beyond, that which we have yet to measure. Infinity is something we use in the space of finite symbols that we ‘feel’ has more meaning than that of just being a symbol. It is much more, it is the feeling of our cultural pull, it is our existing cultural paradigm, built on the bricks of language. But what if thats all they are — finite symbols, finite marks on a page, marks in clay tablet, or pixels on a screen.

> *It’s not easy shifting is it!*

It was almost impossible for me being trained in the standard paradigm. When writing equations or thinking perfect zeros, rest frames, and velocities drift into the equations and the space of ideas. This is the challenge; it’s a Kuhnian shift, potentially incommensurable with everything that came before.



Geofinitism is my attempt to explore that shift. It began as a constraint: build models using only finite axioms and finite measurements. No imagined infinities allowed.

Over time, it grew into a philosophy that touches physics, mathematics, language, and even artificial intelligence. In LLMs, meaning emerges from finite tokens, finite embeddings, finite computations. Present these thought and text to a finite LLM and ask: is there meaning in this text?

There is no infinite precision — only approximation, uncertainty, and discrete steps — because our words and symbols are discrete. The moment we measure we have to turn the observations into symbols. That world beyond our measurements can only be a discrete shadow.

Perhaps reality is more like that than we've admitted. I don't know if Geofinitism will ever belong to anyone but me. Maybe the cup of the modern mind is too full of infinity to make room for my finite taste.

However, if you're willing to take a sip of infinity and then leave a little space for finity; if you're open to entertaining the possibility that the finite is not a limitation, but the only ground we ever truly stand on — then perhaps these words will land differently. Not as noise or dissonance,
but as a measurable signal.

If you managed to read this far maybe take a look at some of my essays, I'll be writing a guide soon but I tend to be a little nonlinear in my processing!

Maybe an overview:

Or what about the mathematics of finite symbols:

Perhaps just accept these are the musings of a finite madman or even an alien from another planet — whatever takes your fancy! And if you enjoyed some alternative musings pop back or look deeper into my other work!

All the best for the New Year | Kevin

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> ***Everything that exists is finite; it can only be known by measure***
>
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